

DEEPAK LENKA

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RESEARCH INTERESTS:

Reasoning, Large Language Models, Pretraining, AI Scientific Discovery, AI Alignment, Interpretability, Graph Structure

Experience:

Intellicuria AI, India — *AI Engineer/Researcher*

October 2023 - January 2024

- I contributed to the development of an innovative image captioning system, directing the fusion of Vision Transformer (ViT) and GPT-3 for detailed image descriptions.
- I meticulously fine-tuned ViT to achieve optimal image processing precision and elevated GPT-3 for the seamless generation of coherent textual descriptions.
- Leveraged GPT-4 for automated radiology image analysis and report generation, enhancing visualization through DICOM to PNG conversion. Streamlined the diagnostics process by seamlessly integrating text-based reports with their corresponding images.

PROJECTS:

[Radiology AI](#) — *Python, Hugging Face Transformers, ViT*

- Implemented AI-driven automation for radiology image analysis and report generation using GPT-4. Orchestrated DICOM to PNG conversion for enhanced visualization and seamless integration of text-based reports with corresponding images. Streamlined radiology diagnostics, showcasing expertise in computer vision and language processing.
- Revolutionized radiology diagnostics through AI-driven automation, leveraging GPT-4 for efficient image analysis and report generation. Achieved a substantial decrease in report generation time, optimizing operational workflows and ensuring prompt patient care.
- It's going to impact radiology reporting, the project empowers physicians with enhanced insights into scan images, fostering quicker and more nuanced understanding. Its profound impact extends to billions of individuals, marking a significant stride in advancing medical diagnostics and patient care.

[WhisperSync](#) — *Whisper ASR , VTT (WebVTT)*

- Implemented an automated phone call transcription system with the Whisper ASR model, segmenting and transcribing caller and callee audio segments. Enhanced accessibility and analysis of spoken content by generating WebVTT-formatted transcriptions and organizing data in JSON format.

[FusionFlow](#) — *llama-index, google-generativeai, qdrant_client*

- The project revolves around GeminiMultiModal, a cutting-edge Large Language Model (LLM), employing llama-index and google-generativeai for efficient language processing and generative AI. It integrates with qdrant_client for vector similarity search and utilizes matplotlib for data visualization. The primary objective is to enhance multimodal completion tasks, particularly in interpreting and generating context from images through advanced language modeling.

[Imagineify AI](#) — *StableDiffusion, Gemini, Python*

- Developed an integrated system utilizing GeminiPro, an advanced generative model, to craft text prompts for Stable Diffusion. Orchestrated an accelerated image generation pipeline, incorporating sd-xl-turbo models, on a CUDA-enabled GPU. Demonstrated expertise in leveraging cutting-edge technologies to facilitate creative content generation through innovative and efficient workflows.

INDUSTRY EXPERIENCE:

Indian Institute of Remote Sensing, Indian Space Research Organisation(Department of Space) June 2023- August 2023

Application of Machine Learning in Urban Studies Student Researcher

- machine learning projects focusing on LULC mapping, building footprint extraction, greenspace identification, urban growth modeling, and slum mapping within the Indian Institute of Remote Sensing.
- Worked closely with interdisciplinary teams to analyze extensive datasets, extract relevant features, and develop predictive models aimed at facilitating sustainable urban development.
- Contributed to the development and delivery of a comprehensive one-week online course on "Machine Learning Application in Urban Studies," covering key topics such as AI/ML techniques, feature extraction, and urban modeling.

Skills:

Languages —

- I primarily code in Python, Java. Some JS, swift C & C++

Technologies—

- Utilized OpenAI GPT-3.5 and 4, pandas, NumPy, Matplotlib, requests, pydicom, Pillow, scikit-learn, TensorFlow, Torch, torchvision, OpenAI Whisper ASR, llama score (llama-index), Google Generative AI (gemini-pro), diffusers, accelerate, and mediapy for diverse projects.

Concepts—

- Possess expertise in Natural Language Processing (NLP) with OpenAI GPT-3.5, data manipulation using pandas, scientific computing with NumPy, data visualization using Matplotlib, web interactions with requests, medical image analysis using pydicom, image processing with Pillow (PIL), machine learning with scikit-learn, deep learning with TensorFlow and Torch, audio transcription with OpenAI Whisper ASR, information retrieval with llama score (llama-index), generative content creation with Google Generative AI (gemini-pro), and GPU acceleration using diffusers and accelerate. Additionally, proficient in media manipulation with mediapy.

GenerativeAI—

- Deep Learning Architectures RNN, LSTM, GAN, Autoencoder.
- Understand Fine Tuning, Zero Shot, Few Shots, RAG .
- Understand Optimization LORA, QLORA, PEFT, RLHF, Supervised Fine Tuning, Prompt Engineering, Langchain, VectorDB, AutoGPT.
- Research Paper.

Hobbies—

- Regarding books, coding, talking to smart people and understanding the universe from a physics point of view.

EDUCATIONS:

Harvard University SC50, Artificial Intelligence(2023)

Completed Harvard’s CS50: Introduction to Artificial Intelligence to Python

Harvard University CS50, Computer Science(2021)

Completed Harvard’s CS50: Introduction to computer science

Godavarish Mahavidyalaya, India— Math and Physics

Collage in Science (Chemistry, math and physics) 2017-2019

COURSES:

Undergraduate Level Courses	Graduate Level Courses
• Intro to Differential Geometry • Machine Learning for Data Sciences • Honor Linear Algebra • Developmental Psychology • Biopsychology of Learning & Memory • Numerical Analysis	- Computer Vision - Foundation of Reinforcement Learning - Statistical Distance for Modern Machine Learning - Analysis of Algorithms - Algorithmic Game Theory - Advance Compilers